

# AI-Powered HMS Synthetic Clinical Dataset

## Lab Result

|                    |                |            |                         |
|--------------------|----------------|------------|-------------------------|
| Patient Name       | Pham Hong Yen  | MRN        | MRN-0021                |
| DOB                | 2003-04-02     | Gender     | Female                  |
| Encounter Date     | 2023-12-24     | Department | Obstetrics & Gynecology |
| Attending Provider | Dr. Dev Doctor | Status     | active                  |

Synthetic Data Warning: This document is generated for software testing, OCR parsing, chunking, embedding, access-control validation, and Graph RAG demonstrations. It is not valid medical advice and must not be used for patient care.

## Lab Report Summary

Six-month lab trend reviewed for Pham Hong Yen with emphasis on renal function, glycemic control, BNP, liver enzymes, electrolytes, and hemoglobin. Clinical context: prenatal or gynecologic follow-up with anemia and glucose screening needs.

### Recent Lab Snapshot

| Date       | Analyte    | Value | Unit          | Reference Range | Status |
|------------|------------|-------|---------------|-----------------|--------|
| 2026-05-01 | Creatinine | 1.29  | mg/dL         | 0.7-1.3         | Normal |
| 2026-05-01 | eGFR       | 68.0  | mL/min/1.73m2 | >=60            | Normal |
| 2026-05-01 | Glucose    | 110.0 | mg/dL         | 70-99           | High   |
| 2026-05-01 | HbA1c      | 5.8   | %             | <5.7            | High   |
| 2026-05-01 | BNP        | 313.0 | pg/mL         | <100            | High   |
| 2026-05-01 | AST        | 30.0  | U/L           | 10-40           | Normal |
| 2026-05-01 | ALT        | 41.0  | U/L           | 7-56            | Normal |
| 2026-05-01 | Potassium  | 4.3   | mmol/L        | 3.5-5.0         | Normal |
| 2026-05-01 | Hemoglobin | 12.0  | g/dL          | 12.0-17.5       | Normal |

## Interpretation

Abnormal values should be correlated with symptoms, medications, hydration status, and department-specific care goals. High BNP or low eGFR should trigger clinician review before medication changes.

# **Trend Notes For AI Extraction**

## **Renal Labs**

Creatinine and eGFR are included for renal dosing and medication safety retrieval. Monitor K when ACEI, ARB, spironolactone, or diuretics are active.

## **Medication Safety Linkage**

The lab trend is intended to link to medication\_safety and renal\_labs retrieval scopes. Pharmacist review is recommended for high-risk drug-lab combinations.